UNIT
2

GASEOUS EXCHANGE IN HUMANS

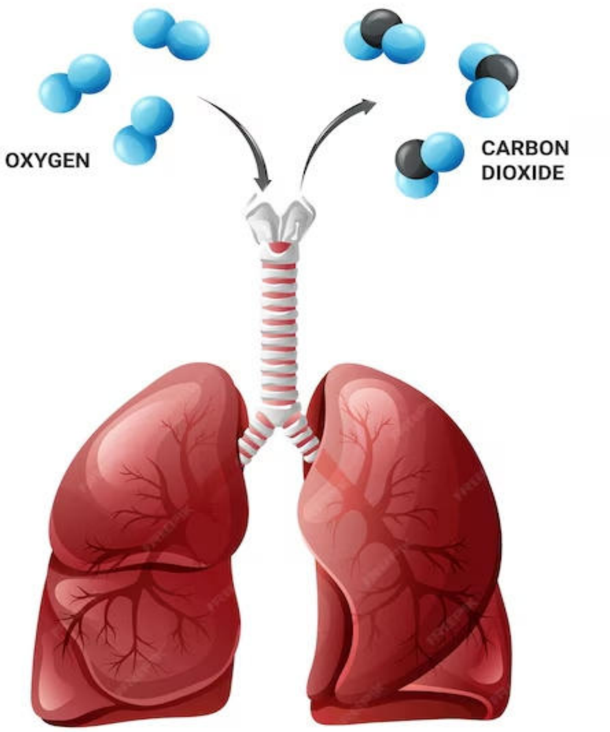
Learning objectives:

- Explain the process of respiration and its role in energy production.
- Identify the components of the human respiratory system and their functions.
- Understand the process of gas exchange in the lungs and its importance for survival.

Respiration is the process of obtaining energy by breaking down glucose, releasing carbon dioxide and water as waste. Gaseous exchange allows organisms to take in oxygen and expel carbon dioxide. This chapter explains how the human respiratory system supports gaseous exchange.

2.1 What is Respiration?

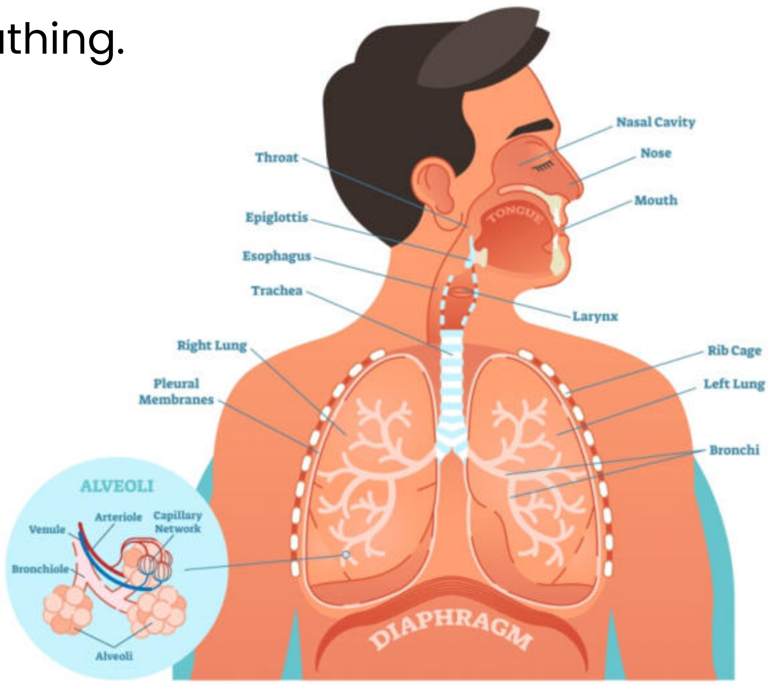
Respiration provides energy for all the activities in our body, such as movement, growth, and repair. It involves taking in oxygen from the air and releasing carbon dioxide. Oxygen is used to break down food molecules, producing energy.



2.2 Components of the Human Respiratory System

The human respiratory system consists of:

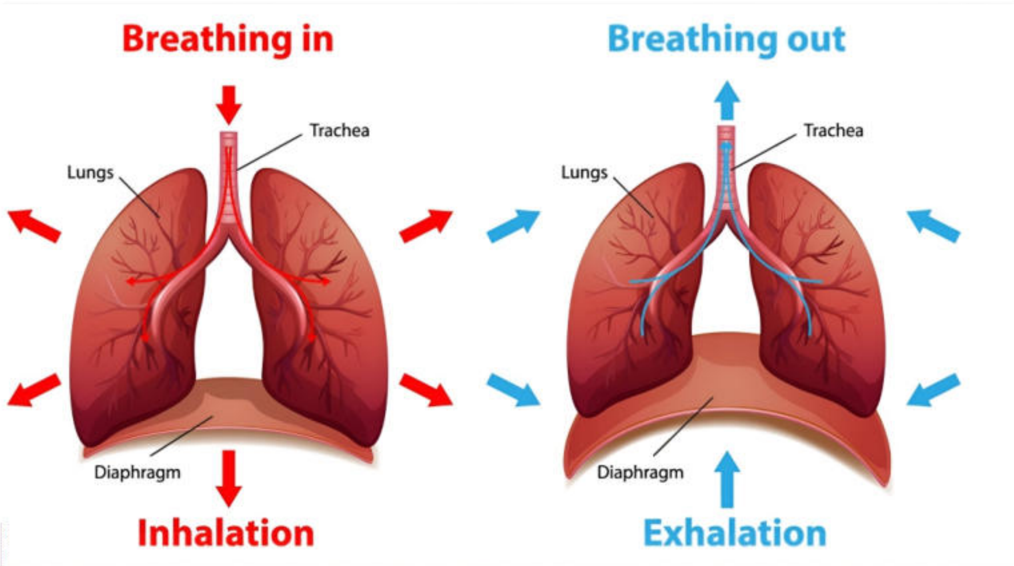
1. **Nasal Cavity (Nose):** It filters, warms, and moistens the air.
2. **Trachea:** A windpipe that carries air to the lungs.
3. **Bronchi:** They are the branches of the trachea that lead to each lung.
4. **Lungs:** The main organs for gaseous exchange.
5. **Alveoli:**They are tiny air sacs in the lungs where oxygen and carbon dioxide are exchanged.
6. **Diaphragm:** A muscle under the lungs that helps in breathing.



2.3 The Process of Breathing

- **Inhalation:** The diaphragm contracts and moves down, creating space in the chest, and air enters the lungs. Oxygen is taken in during inhalation.

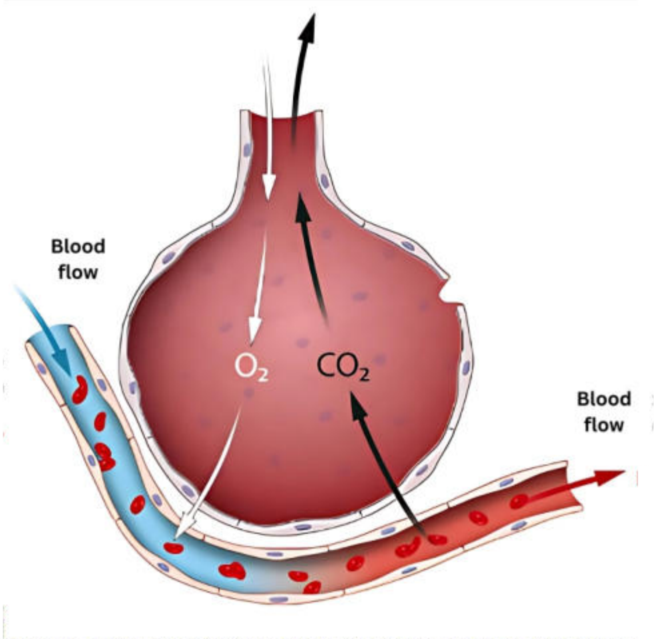
- **Exhalation:** The diaphragm relaxes, reducing chest space, and air is pushed out of the lungs. Carbon dioxide is released during exhalation.



2.4 Gas Exchange in the Lungs

Gas exchange occurs in the alveoli:

1. Oxygen passes from the alveoli into the blood.
2. Carbon dioxide from the blood moves into the alveoli to be exhaled.



2.5 Transport of Gases in Blood

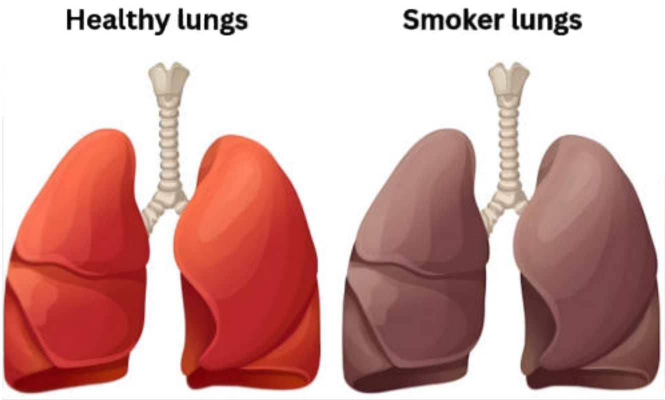
- **Oxygen:** It is carried by red blood cells to all body parts.
- **Carbon Dioxide:** It is carried from body tissues to the lungs for exhalation.

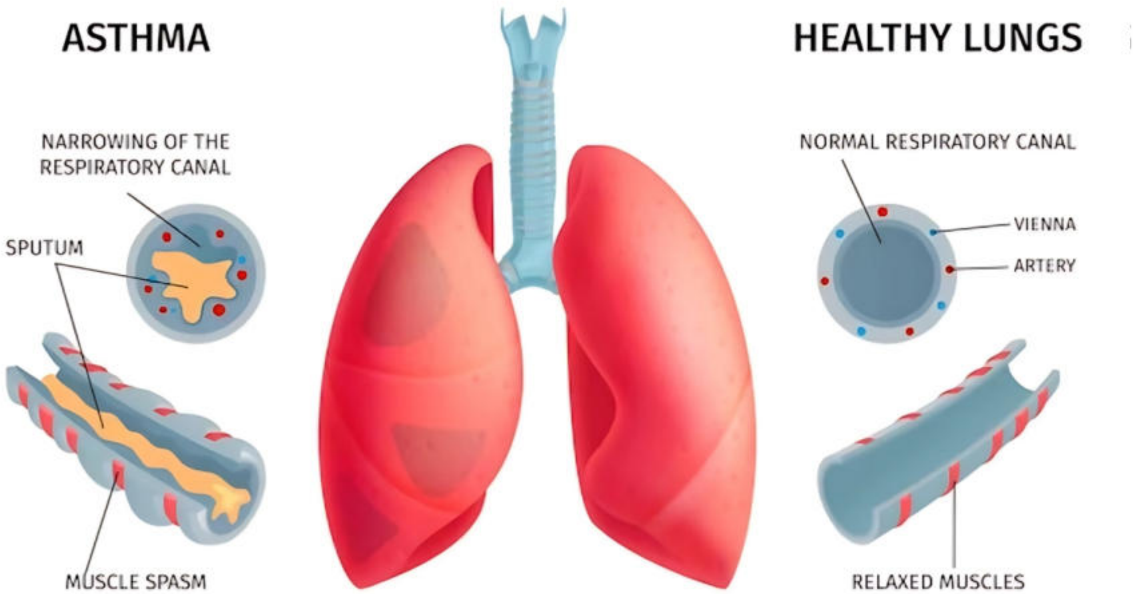
2.6 Common Respiratory Disorders

Respiratory disorders can interfere with the efficient exchange of gases. Here are some common disorders:

- **Asthma:** It represents by narrowed airways cause wheezing, coughing, and shortness of breath. Triggers include dust, pollen, and smoke.
- **Bronchitis:** It indicates inflammation of bronchial tubes leads to coughing and mucus production. Caused by infections or irritants like smoke.

- **Pneumonia:** The type of infection that fills alveoli with fluid, causing chest pain, fever, and breathing difficulty.
- **Tuberculosis (TB)**: It is a bacterial infection that spreads through droplets, causing persistent cough, fever, weight loss, and night sweats.
- **Smoking-Related Diseases**: Smoking damages lungs, leading to COPD (Chronic obstructive pulmonary disease) and lung cancer.





EXERCISES

Fill in the blanks.

1. The ______________filters, warms, and moistens air entering the body.

2. Oxygen is absorbed into the blood in the ______________.

3. The ______________ carries air from the nose to the lungs.

4. The main organs for gaseous exchange in humans are the ______________.

5. ______________ is a respiratory disorder caused by narrowing of airways.

6. In pneumonia, the ______________ fill with fluid.

7. ______________ is carried by red blood cells.

8. ______________ damages lung tissue and can lead to cancer.

Tick(✓) for True and cross (x) for False statements.

1. The lungs are the main organs for gaseous exchange. ☐

2. Asthma is caused by bacteria. ☐

3. Smoking can lead to lung cancer. ☐

4. Gas exchange takes place in the alveoli. ☐

5. Pneumonia causes the alveoli to fill with fluid. ☐

6. Bronchitis affects the diaphragm. ☐

7. Tuberculosis is caused by a virus. ☐

8. The nose plays no role in respiration. ☐

Choose the correct answer.

1. Gas exchange occurs in the:

a) Trachea
b) Alveoli
c) Diaphragm
d) Bronchi

2. The waste gas produced in respiration is:

a) Oxygen
b) Carbon dioxide
c) Nitrogen
d) Methane

3. It helps in breathing by moving up and down:

a) Heart
b) Diaphragm
c) Bronchi
d) Lungs

4. Oxygen is carried in the blood by:

a) Platelets
b) White blood cells
c) Red blood cells
d) Plasma

5. The tiny air sacs in the lungs are called:

a) Trachea
b) Bronchioles
c) Alveoli
d) Capillaries

6. Smoking damages the:

a) Heart
b) Lungs
c) Liver
d) Kidneys

7. Tuberculosis is caused by:

a) Virus
b) Bacteria
c) Fungi
d) Parasites

8. Asthma can be triggered by:

a) Dust
b) Bacteria
c) Cold water
d) Smoke

Answer the Following Questions:

Short-Answer Questions

1. What is respiration?
2. Name the main organs of the human respiratory system.
3. Name two respiratory disorders.
4. What are the symptoms of pneumonia?
5. How does smoking affect the lungs?

Long-Answer Questions

1. Write the difference between inhalation and exhalation.
2. Explain the role of alveoli in gaseous exchange.
3. Write a note on asthma and its causes.
4. Discuss the effects of smoking on the respiratory system.
5. Explain the components of human respiratory system?
6. Draw a labelled diagram of human respiratory system?

Short Note

1. Write a short note on common respiratory disorders.

Answer the questions using the words provided in the box.

oxygen	involuntary	carbon dioxide

a) What gas does a human being take in when we breathe?

We inhale _______________.

b) What gas do human beings eliminate during respiration?

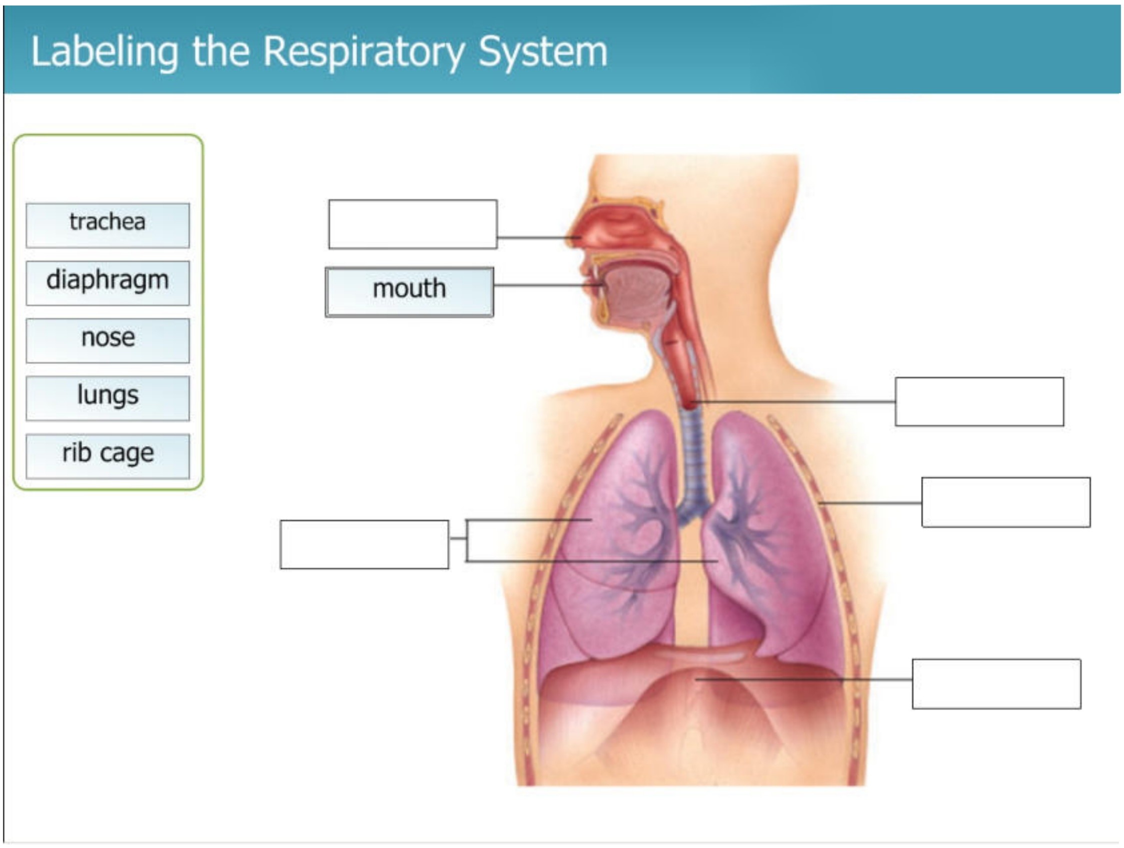
We exhale _______________.

c) Is respiration a voluntary or involuntary movement?

It is an _______________ movement.

1.2. Name the organs of the respiratory system:

Label the diagram, match the names listed above.



Comparing Breathing Rates

ACTIVITY

Objective: Measure how breathing changes during rest and after exercise.

Materials: Stopwatch, notebook, and volunteers.

Instructions:

1. Count the number of breaths a person takes in one minute while resting.
2. Perform a simple exercise (e.g., jumping jacks) for 2 minutes.
3. Count the number of breaths again.
4. Compare the results.

Expected Outcome: Students observe an increase in breathing rate after exercise.

Critical Thinking

Why do you think people with asthma find it harder to breathe during an attack compared to healthy individuals?